

RAM BODEPUDI

+1(972)510-8164 | Dallas, TX | rammaruthyb@gmail.com | [Ram Maruthy LinkedIn](#) | <https://github.com/ram5112>

OBJECTIVE

An accomplished AI Engineer with 4 years of experience delivering cutting-edge AI solutions across diverse industries. Proficient in developing and deploying machine learning models, generative AI applications, and scalable data pipelines using frameworks like Hugging Face Transformers, TensorFlow, and PyTorch. Experienced in implementing NLP-driven solutions, predictive analytics, and conversational AI to enhance operational efficiency and user engagement. Skilled in leveraging cloud platforms like AWS for deploying robust, scalable AI systems and automating workflows with Docker and CI/CD pipelines. Certified AWS Developer Associate with expertise in Python, data engineering, and AI integration, consistently driving impactful, data-driven results.

EDUCATION

Masters of Computer Science , University of Texas at Arlington	2022 - 2024
Bachelor of Computer Science , Gandhi Institute of Technology and Management	2017 - 2021

SKILLS

Programming Languages: Python, C++,Java,SQL, Javascript, Html, Css, R	Data Analysis: Pandas, NumPy, Matplotlib, Seaborn
Machine Learning: TensorFlow, PyTorch, Keras, scikit-learn,MLLib,MLFlow,Theano, Caffe	Integration and Reporting: SSIS, PowerBI, Tableau
Gen Ai and Deep learning Models: LLM-Langchain , Llama, Transformers(BERT, Huggingface)	Database: SQL Server, MySQL, PostgreSQL, Sqllite
Frameworks: Flask, Django, React,Streamlit	Feature Store: Hopsworks, AWS SageMaker
	Big Data and Cloud: AWS,Azure,GCP, Terraform
	Version Control: Git, Github
	Deployment and Scaling : Docker, Kubernetes

EXPERIENCE

Humana TX, USA

Gen AI Engineer	Jun 2024 - Present
- Leveraged Hugging Face Transformers to implement Retrieval-Augmented Generation (RAG), integrating pre-trained language models with a robust retrieval mechanism to enhance the chatbot’s ability to generate accurate and context-aware responses across various domains, including customer support and sales. - Integrated Neural Networks and SVM as part of the RAG pipeline to refine user intent classification. Designed and optimized scalable retrieval pipelines using AWS S3 and EC2 to handle large volumes of user interactions efficiently. These pipelines ensured low-latency access to relevant knowledge documents during real-time query resolution. - Applied advanced NLP techniques such as Named Entity Recognition (NER) and sentiment analysis to preprocess and categorize user queries, enabling the chatbot to personalize responses based on user context, sentiment, and preferences. - Developed fine-tuned transformer-based models using Hugging Face and Google Dialog Flow to empower the chatbot with context-aware retrieval capabilities. These models excelled at handling complex, multi-turn conversations by dynamically querying and integrating relevant external knowledge into the generated responses. - Engineered ensemble models with decision trees and boosting methods (e.g., XGBoost) for improving retrieval ranking. These models effectively prioritized the most relevant knowledge snippets, reducing errors in response generation and enhancing user satisfaction. - Collaborated with cross-functional teams to integrate the RAG-based chatbot into existing customer support frameworks. This integration streamlined operations and achieved a 15% reduction in resolution times by providing precise, contextually enriched answers. - Automated data preprocessing and retraining workflows using AWS Lambda and EC2, enabling the RAG models to maintain relevance by incorporating the latest information into their retrieval components. - Developed personalized retrieval and generation mechanisms to adapt chatbot interactions based on user history and behavior. This approach increased user satisfaction and engagement by delivering highly relevant, knowledge-backed responses.	

Environment: Python, Hugging Face Transformers, PyTorch, TensorFlow, Jupyter Notebook, VSCode, Scikit-learn, Keras, NumPy, Matplotlib, AWS SDK (Boto3), AWS Lambda, S3, EC2, Pandas, SpaCy, NLTK, BERT, Hugging Face Pipelines, AWS Comprehend, TensorFlow, Dialog Flow API, REST APIs, Flask/Django, Postman, XGBoost, Random Forest, Elasticsearch.

Cargill, Hyderabad, India

Machine Learning Engineer

Jan 2021 - June 2022

- Developed machine learning models using Scikit-learn and Pandas to predict truck delays based on historical route data, weather conditions, and traffic patterns, improving delivery time accuracy and logistics efficiency.
- Built scalable data pipelines for ingestion, cleaning, and transformation using Pandas, Airflow, Hopsworks, and Apache Spark on Amazon EMR to accommodate continuous training on new data.
- Migrated backend systems from PostgreSQL to Amazon RDS, ensuring scalability and seamless integration with existing infrastructure.
- Deployed prediction models using Docker on Amazon ECS, enhancing scalability and operational efficiency for production environments.
- Automated CI/CD pipelines with GitHub Actions and AWS CodePipeline, reducing manual intervention and ensuring consistent deployments and model updates.
- Integrated MLflow for experiment tracking, model versioning, and reproducible workflows, improving collaboration and transparency.
- Created an interactive dashboard using Streamlit, deployed on Amazon Fargate, EC2 using ECS and EKS, to monitor and visualize truck delays and predicted arrival times for logistics managers.
- Conducted data preprocessing using NumPy and Pandas, ensuring high-quality inputs from GPS, weather APIs, and traffic data for accurate predictions.
- Developed RESTful APIs with Flask to provide truck delay predictions to third-party logistics platforms and integrated real-time traffic and weather APIs for enhanced model accuracy.
- Utilized Hopsworks and MLflow for feature engineering and data versioning, improving efficiency and reproducibility in feature extraction for machine learning workflows.

Environment: Python, SQL, Scikit-learn, Pandas, NumPy, MLflow, Matplotlib, Seaborn, Hopsworks, Feast, AWS Lambda, PostgreSQL, Amazon RDS, Amazon ECS, Docker, AWS EC2, AWS CodePipeline, GitHub Actions, Flask, RESTful APIs, third-party traffic and weather APIs, Streamlit, Jupyter Notebook, Vscode, Spark.

Fractal Analytics, Hyderabad, India

Data Scientist

Apr 2019 - Dec 2020

- Analyzed clinical trial data using Python, Pandas, and advanced SQL techniques to identify key patterns, optimize stored procedures, and provide actionable insights that enhanced research outcomes and decision-making.
- Built machine learning models with Scikit-learn and TensorFlow to predict outcomes in personalized medicine and implement predictive maintenance strategies, improving accuracy and operational efficiency across use cases.
- Designed scalable, real-time data pipelines with PostgreSQL and AWS services like Amazon EMR, EC2, and S3, enabling seamless data ingestion, cleaning, and processing to support continuous model training.
- Automated data workflows using SSIS, integrating Python scripts with libraries like Beautiful Soup, PyPDF2, and pdf plumber to streamline data extraction from web pages and PDFs, improving efficiency and accuracy.
- Developed advanced Power BI dashboards and Matplotlib visualizations to automate reporting, provide real-time insights for stakeholders, and enable dynamic, interactive exploration of multi-source datasets.
- Conducted natural language processing (NLP) on medical literature to support drug development, extracting relevant insights while leveraging web scraping tools to acquire and preprocess valuable data from diverse sources.
- Integrated machine learning models into healthcare applications using Flask APIs, ensuring secure access to predictions, while incorporating MLflow for tracking experiments, versioning models, and monitoring performance metrics.
- Optimized database performance through advanced SQL indexing techniques and implemented predictive algorithms for supply chain management, improving scalability, query performance, and data-driven decision-making capabilities.

Environment: Python, Pandas, NumPy, Scikit-learn, TensorFlow, PostgreSQL, Amazon EMR, AWS EC2, AWS S3, Matplotlib, Flask, MLflow, GitHub, Jupyter Notebook, Visual Studio Code, SQL, Power BI, SSIS, Beautiful Soup, Requests, PyPDF2, pdf plumber, Docker, Linux, Tableau, Apache Spark, and PyTorch.

CERTIFICATIONS

AWS Certified Developer Associate

Python Certified Programmer [PCEP] and IBM Applied AI Professional Certification